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Poster

Distributed Power System for Microsatellites
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Abstract

The Aerospace Corporation has independently developed a distributed "Ring Bus" electric power system for Picosatellites. This distributed power system architecture is being implemented on a number of upcoming Picosatellite missions. The first implementation of this architecture will be flown on the PowerSphere Flight Experiment and the Pico Satellite Inflatable Reflector Experiment (PSIREX).

To date all of the Picosatellite missions do not have attitude control or deployable solar array structures. Solar cells are body mounted on the various sides of the Picosatellite. The "Ring Bus" architecture was conceived to solve the problem of obtaining the maximum amount of electric power from a solar array, that has multiple panels, that are not arranged on a single planar surface.

Aerospace Corporation researchers working on developing viable power systems for Picosatellites were awarded patents 6,127,621 October 2, 2000 "Power Sphere" and 6396167 B1 May 28, 2002 "Power Distribution System" for this unique solution for distributed power system architecture for a multifaceted solar array.

The authors have developed a prototype of the ring bus and have completed initial testing of its performance.

PowerSphere Flight Experiment

The PowerSphere is a Picosatellite bus, which will provide NASA and DoD users with an effective, flexible approach for communications, inspection, sensing, and surface surveying.

The PowerSphere development (begun March 2001, under NASA contract NAS3-01115) is a collaborative effort with team members from The Aerospace Corporation, ILC Dover, Lockheed Martin, and NASA GRC. This technology development effort has been completed with the fabrication and testing of an engineering development unit shown in Figures 1-3.

The PowerSphere is a highly capable spacecraft bus with attitude-insensitive power generation and multi-beam redundant communications systems capabilities that could be used on a variety of future NASA and DoD missions. Potential missions for these small satellites could include communications relays in a space-based wide area network. They can also be deployed for autonomous surveillance of a manned spacecraft, and on-

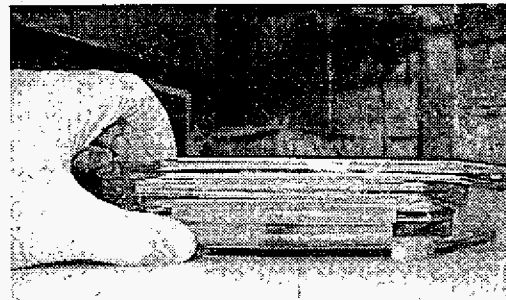


Figure 1. Stowed PowerSphere Hemisphere

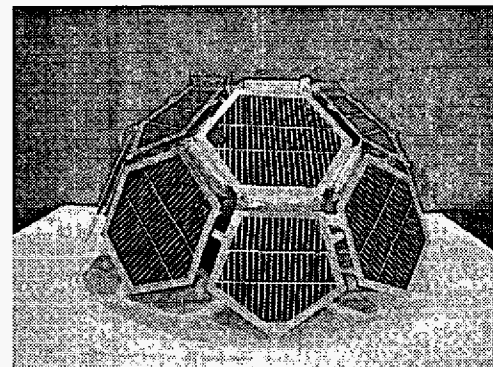


Figure 2. Deployed PowerSphere Hemisphere

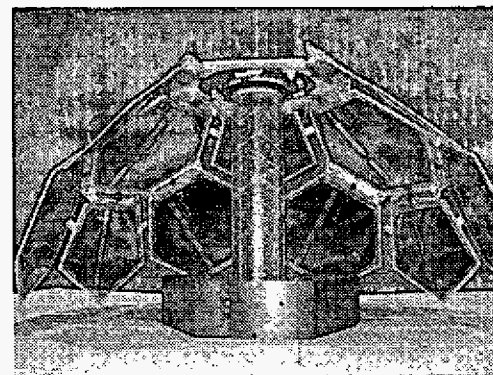


Figure 3. Cutaway view of Deployed PowerSphere Hemisphere

orbit monitoring of Lunar and Mars exploration vehicles. Another application would be a mini Global Positioning System (GPS) constellation for surface coverage of the Moon or Mars. The use of these extremely small satellites will allow NASA to build and deploy the basic infrastructure that is needed for robotic and human exploration of the solar system at an affordable cost, in a highly reliable manner that is effective and flexible to changing requirements.

A PowerSphere with a deployed diameter of 0.6 m and >10 kg can supply 10 watts orbit average of power. A PowerSphere with electric thrusters would be capable of 60 m/s delta over the mission life. This maneuvering capability would enable long-duration inspection of host spacecraft or for station keeping for a communication relay satellite.

PSIREX Flight Experiment

The PSIREX flight experiment is designed to provide the United States Air Force with space flight data, that will be used to validate performance models for large inflatable structures and solar collectors. The experiment will stow a one-meter diameter thin film optical reflector in a space, that is 5 inches by 5 inches by 2.5 inches. The total size of the PSIREX Picosatellite is 5 inches by 5 inches by 10 inches. The satellite will have four Emcore ATJ solar cells mounted on each of the four sides, which have dimensions of 5 inches, by 10 inches. Figure 4 shows the engineering development unit PSIREX Picosatellite body with the Emcore Solar Cells mounted on it. Figure 5 shows the orbital configuration of the PSIREX with the reflector deployed.

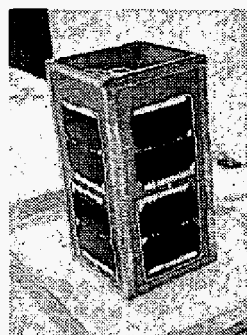


Figure 4. Photograph of PSIREX Spacecraft Body with Emcore ATJ Solar Cells

The PSIREX Picosatellite will have an orientation during flight, that will keep the side of the spacecraft body perpendicular to the direction of flight. This orientation will result in a less than optimal orientation of the solar cells on the spacecraft body during each orbit. In addition, it is expected that the spacecraft will spin at some undetermined rate around this axis. Figure 6 provides a simulation of the power producing capability of the PSIREX Solar array.

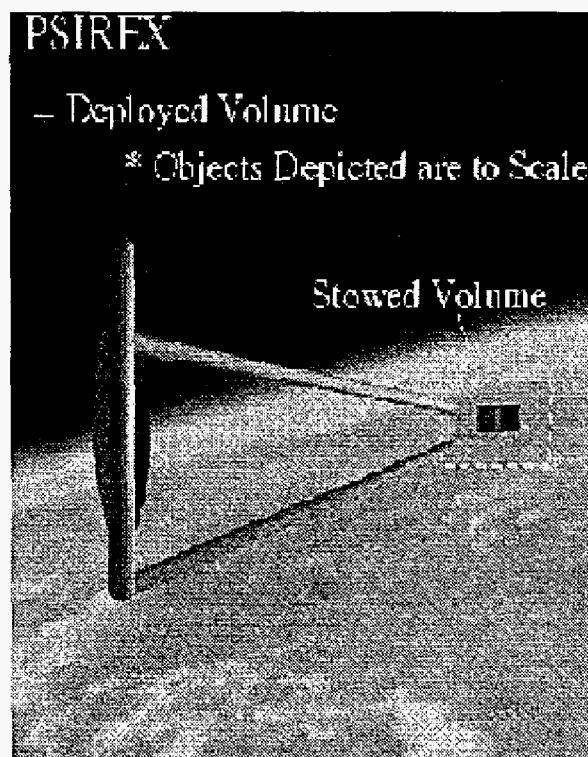


Figure 5. Artist rendition of PSIREX with one-meter Inflatable reflector deployed in low Earth orbit

Description of Operation of PMAD

The PowerSphere and PSIREX Power Management and Distribution (PMAD) systems utilize the "Ring Bus" power system architecture. This architecture was originally developed for the PowerSphere. The configuration of the major elements of the PMAD for the Ring Bus is shown in Figure 7. In this configuration each of the individual power modules sense the voltage on the "Ring Bus" which is set nominally at 5 VDC for PowerSphere or 10 VDC for the PSIREX Picosatellite. The PSIREX ring bus is more mature at this juncture and will be discussed first.

The Solar array regulators each have a microprocessor and a DC-DC converter. The solar array on each face of the PSIREX has the four solar cells connected two in series with the two series strings connected in parallel. Two of these arrays on opposing faces of the PSIREX spacecraft are connected to a single solar array regulation unit through blocking diodes. For operation in the sunlight portion of the orbit the first level of the control loop is the Pulse Width Modulated (PWM) DC-DC boost converter, which provides a regulated bus with voltages between 9.5 and 10.5 volts. If left to alone this PWM DC-DC converter would increase the current demand on the controlled solar array beyond the peak

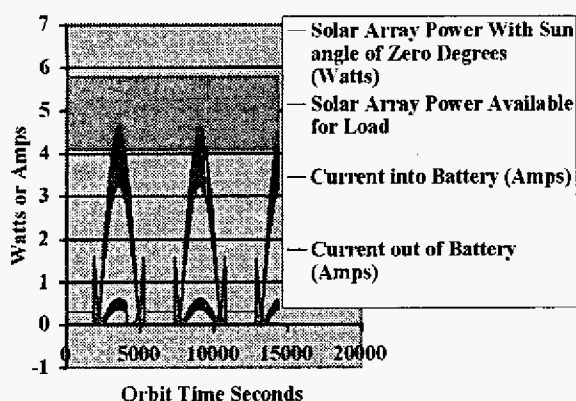


Figure 6. Power producing capacity of the PSIREX Solar Array

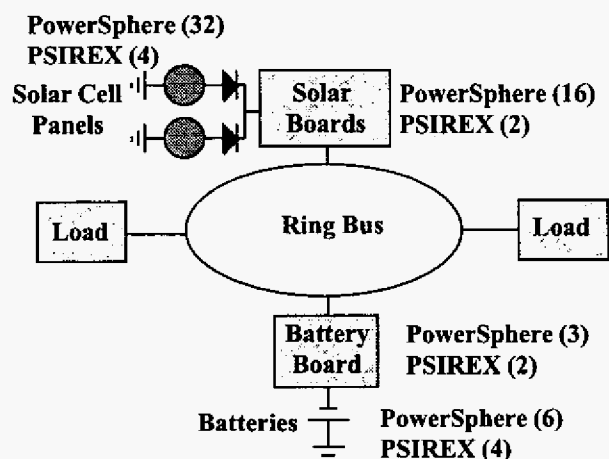


Figure 7. Schematic Diagram of Ring Bus Architecture

power point. If this happens the power output of the converter would collapse to zero. To prevent this from happening a microprocessor monitors the bus voltage and output current and implements a peak power-tracking algorithm. Thus if an increase in current demand from the PWM DC-DC converter results in a decrease in power output the microprocessor commands a lower current demand by the PWM DC-DC converter. The microprocessor also monitors the solar array voltage and turns the PWM DC-DC converter off if the solar array voltage drops below 3.0 volts and turns it back on when the voltage exceeds 3.2 volts. The battery subsystems provide power to the "Ring Bus" when the bus voltage drops below 10.0 volts. When the bus is supported by the batteries the battery microprocessors turn off all of the battery chargers, sets a flag in memory to swap the battery to be charged when in sunlight again and goes to a low power sleep state. The basic building block for the battery subsystem is a battery control element, that controls the operation of two individual battery cells and their associated chargers, and boost regulators. During the sunlight portions of the orbit the battery building block microprocessor turns off one of the battery boost

regulators and turns on the associated battery charger. As the bus voltage increases, the microprocessor allows the battery charge current to increase slowly to a maximum of 0.75 amps. If the bus voltage decreases due to an increase in load or decrease in solar illumination, the battery charger immediately decreases the charge current to the maximum amount available after supplying all other loads. If the bus voltage drops below 10.1 volts the charger is turned off and the boost regulator is turned back on.

Test Results

A Representative Ring Bus for the PowerSphere was assembled with two solar array regulators and two battery charger/regulators. The overall system efficiency was measured at two different load conditions. The results of this test are presented in Table 1.

The schematic for the PSIREX solar array regulator board is shown in Figure 8. A test was performed on this board and the operating efficiency was measured as a function of solar array input voltage with two different loads. The results of this test are presented in Table 2.

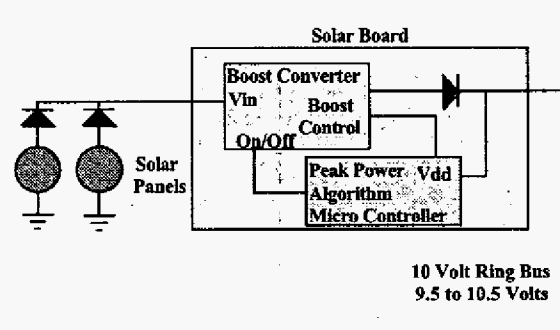


Figure 8. Schematic of PSIREX Solar Array Regulator Board

Conclusions

The Ring Bus developed for this small satellite solves the problem of maximizing the outputs of multiple solar array panels each of which may be at different temperatures and have different orientations with respect to the sun. The voltage of the ring bus should be selected to directly power the satellite's largest load. The solar array controller must have a peak-power-tracking algorithm to automatically provide peak power output of the solar arrays to the Power Bus under all conditions.

Acknowledgements

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Table 1. Efficiency Measurements of PowerSphere Ring Bus

5 Volt Ring Bus with Solar Board (Rev B -no mods) and Battery Bd (Rev B with charger mods) Efficiency Test										
2 Solar source inputs and 2 Battery inputs (all current metered)										
Solar Input watts = SA1V*SA1C+SA2V*SA2C										
Load Watts =BAT1V*BAT1C + BAT2V*BAT2C + (5V)^2/RLoad + 5V*(the 2*PIC's current in non sleep mode)										
Note: 5V*(the 2*PIC's current in non sleep mode) ~ 5V*12ma = .06 Watt										
Case 1 Battery 2 Charging with RLoad=371 ohm										
Solar Input Data					Battery Data				Load	Efficiency
SA1V	SA1C (ma)	SA2V	SA2C (ma)	(watts)	BAT1V	BAT1C	BAT2V	BAT2C (ma)	watts	(%)
4.8	95	4.8	95	0.91	3.8	0	4.0	130	0.58	63.6
Case 2 Battery 2 Charging with RLoad=47 ohm										
4.5	290	4.4	310	2.67	3.8	0	4.0	80	1.63	61.1
Note: each MAX608 with it's ORing diode is about 75% efficient										
Each MAX1926 is about 90% efficient with a 5 Volt input										
For the above, the BAT1 608 draws no current, the BAT2 608 is off (its charging). The solar 608s are supplying all the current										

Table 2. Efficiency Measurements of PSIREX Solar Array Board

10 Volt Ring Bus Solar Board (Rev B with mods) Load Test						
Case 1 Rload=20 ohm						
Solar (volts)	Solar Current (Amps)	Solar (Watts)	Vbus	I Load (ma)	Load (Watts)	Efficiency (%)
4.0	1.78	7.12	10.60	494	5.24	73.5
3.5	2.04	7.14	10.53	478	5.03	70.5
3.3	2.17	7.16	10.50	475	4.99	69.6
3.2	2.30	7.36	10.40	472	4.91	66.7
3.1	2.35	7.29	10.35	465	4.81	66.1
3.0	Drops Out - Because PIC turns 608 off at 3.0V					
	Bringing V _{solar} Back up to 3.5, V _{Bus} kicks back in to 10.53					
Case 2 Rload=15 ohm						
4.1	2.20	9.02	10.48	637	6.68	74.0
4.0	2.26	9.04	10.43	631	6.58	72.8
3.9	2.32	9.05	10.40	630	6.55	72.4
3.8	2.40	9.12	10.33	627	6.48	71.0
3.7	2.17	8.03	9.70	588	5.70	71.0
3.6	2.17	7.81	9.50	578	5.49	70.3
3.5	2.18	7.63	9.30	568	5.28	69.2
3.4	2.18	7.41	9.20	558	5.13	69.3
3.3	2.19	7.23	9.00	547	4.92	68.1
3.2	2.20	7.04	8.80	536	4.72	67.0
3.1	2.20	6.82	8.60	525	4.52	66.2
3.0	Drops Out - Because PIC turns 608 off at 3.0V					
	Bringing V _{solar} Back up to 3.5, V _{Bus} kicks back in to 9.3					

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